

Global Time Echoes IV

Held-Out Replication in the Public MGEX Combined Multi-GNSS Clock Product, 2025–2026

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The Temporal Equivalence Principle (TEP) predicts correlated phase-coherent disturbances in GNSS timekeeping with a spatial correlation length of order thousands of kilometres, an east–west anisotropy exceeding the north–south counterpart, coupling to Earth orbital velocity, and a preferred axis near the CMB rest frame. Papers 1 and 2 established these signatures in GPS-only precise point positioning (PPP) products from three analysis centres spanning 2000–2025, and Paper 3 reproduced them in raw RINEX single-point positioning (SPP), demonstrating independence from analysis-centre orbit and clock models.

This paper presents a held-out replication using a deliberately independent data product: the public MGEX combined multi-GNSS receiver-clock solution (CODE COD0MGXFIN) distributed by NASA CDDIS, for the held-out window 2025-01-01 to 2026-05-01. Receiver-clock offsets for 256 globally distributed stations are read directly from the daily 5-minute CLK files; no positioning is performed. Because MGEX clock files contain a single combined multi-GNSS solution per station, this test provides temporal and product-type independence from Papers 1–3 rather than per-constellation independence, which those papers already address. All six predictions—correlation length, azimuthal anisotropy, orbital-velocity coupling, CMB-frame alignment, ionospheric independence, and geometric robustness—were frozen before the held-out data were inspected.

Four of the five evaluated signatures are recovered. The isotropic correlation length is $\lambda = 1396 \pm 90$ km ($R^2 = 0.486$, 1.75 million pairs), shorter than the 3,000–5,000 km reported for GPS PPP and consistent with the different metric and combined-clock product. The signal persists on geomagnetically quiet days and all four null controls collapse to negligible structure ($R^2 \approx 0$). The full-range anisotropy is modest (ratio 1.23, $p = 0.48$) but the predicted east–west excess emerges in the longitude-matched subset (ratio 2.28, pair-bootstrap $p = 0.002$). The traditional monthly λ and EW/NS ratio do not correlate with orbital velocity (Bonferroni $p > 0.5$), but a supplementary PA-difference metric that avoids exponential-fit noise recovers coupling ($r = -0.670$, $p = 0.017$). The CMB-frame test detects a significant anisotropy axis (LEE $p < 0.0001$) at RA = 60° , Dec = -60° that lies 92° from the CMB dipole, consistent with an ionospheric origin. A supplementary satellite-clock analysis using SP3 orbit geometry finds no detectable spatial correlation, consistent with the single-reference-time nature of the MGEX combined solution.

Frozen predictions and outcomes (MGEX combined-clock product, held out):

- **Correlation length:** λ in the thousands-of-km range — recovered, $\lambda = 1396 \pm 90$ km ($R^2 = 0.486$).
- **Anisotropy:** EW > NS — full-range ratio 1.23 ($p = 0.48$, not significant under station-clustered bootstrap); longitude-matched subset ratio 2.28 (pair-bootstrap $p = 0.002$, station-clustered $p = 0.244$).
- **Orbital coupling:** monthly λ and EW/NS ratio vs Earth orbital velocity — not recovered via traditional metrics (Bonferroni $p > 0.5$); supplementary PA-difference metric recovers coupling ($r = -0.670$, $p = 0.017$).
- **CMB alignment:** full-sky axis scan with look-elsewhere correction — anisotropy axis detected (LEE $p < 0.0001$) at RA = 60° , Dec = -60° that lies 92° from the CMB dipole, consistent with an ionospheric rather than TEP origin.
- **Ionospheric independence:** Kp stratification and storm-day exclusion — signal persists on quiet days.
- **Geometry robustness:** hemisphere-balanced and distance-matched subsets — λ stable under distance matching.

1. Introduction

The Temporal Equivalence Principle (TEP) posits that the proper-time field acquires a dynamical, environment-dependent component at cosmological densities, screened by environment-dependent Gradient Screening with saturation scale $\lambda_T \approx 20 \text{ g cm}^{-3}$ (Paper 0, §7). In the weak-field, low-velocity limit this produces spatially correlated phase-coherent disturbances in precision timekeeping systems, with a characteristic correlation length λ_T of order thousands of kilometres and a preferred anisotropy axis aligned with the CMB dipole.

Papers 1 and 2 of this series established TEP signatures in GPS-only precise point positioning (PPP) products from CODE, IGS and ESA, spanning 2000–2025. Paper 3 moved to raw RINEX single-point positioning (SPP), demonstrating that the same signatures persist without reliance on analysis-centre orbit or clock products. Between them, those papers already provide multi-centre, multi-decade, and raw-observation independence. The principal forms of robustness they do not yet supply are independence from the historical training epoch and independence from the specific GPS PPP product family.

This paper (Paper 4) addresses those two gaps. The analysis uses the public MGEX combined multi-GNSS receiver-clock solution—the CODE COD0MGXFIN CLK product distributed by NASA CDDIS—for the held-out window 2025-01-01 to 2026-05-01, a period chosen to be strictly disjoint from the data used in Papers 1–3. The MGEX clock product is generated by combining GPS, GLONASS, Galileo and BeiDou observations into a single receiver-clock estimate per station, using software, orbit and combination strategies that differ from the GPS-only PPP of Papers 1–2. It therefore constitutes an independent data product at an independent epoch. Every test—correlation length, EW/NS anisotropy, orbital-velocity coupling, CMB-frame alignment, ionospheric independence, and geometric robustness—was defined before the held-out data were inspected. This is a held-out replication paper, not a discovery paper.

It should be emphasised at the outset what this product can and cannot test. Because the MGEX clock files contain one combined multi-GNSS solution per station, they do not permit an independent per-constellation comparison; that form of independence is the province of raw per-system processing, which lies outside the scope of this combined-product analysis. Recovery of the frozen signatures in this independent product and epoch strengthens the case against epoch-specific and GPS-PPP-specific systematics; a null usefully bounds the original interpretation.

2. Data and Processing

2.1 Data Source

All data are public and freely available from NASA CDDIS (cddis.nasa.gov) and the International GNSS Service (IGS). The analysis uses the Multi-GNSS Experiment (MGEX) combined receiver-clock product COD0MGXFIN produced by the Center for Orbit Determination in Europe (CODE), with the Wuhan (WUM) and CNES (GRG) MGEX products as fallbacks on the few days CODE is unavailable. These are daily clock (CLK) files at a 5-minute sampling interval. Each file reports a single combined multi-GNSS receiver-clock estimate per station, derived by CODE from GPS, GLONASS,

Galileo and BeiDou observations; the combined solution is the quantity analysed. No positioning is performed and no raw RINEX observation files are used.

2.2 Time Span and Selection

The analysis period is 2025-01-01 to 2026-05-01 (473 days with usable data), yielding approximately 16 months of held-out daily files. The window is chosen to be strictly disjoint from the data used in Papers 1–3, ensuring temporal independence. Receiver-clock offsets are extracted from the AR (receiver-clock) records of each CLK file for 256 globally distributed stations, converted to nanoseconds, and assembled into per-station daily time series.

2.3 A Single Combined Solution

Unlike Papers 1–3, which can isolate individual systems, the MGEX CLK product contains one combined receiver clock per station rather than per-constellation clocks. The four constellations therefore cannot be separated in this product, and the per-constellation results reported below are, by construction, identical; they are retained only for reporting compatibility with earlier papers. The independence offered here is temporal (a held-out epoch) and product-type (a combined multi-GNSS clock from a different analysis pipeline), not per-constellation.

2.4 Frozen Prediction Register

Before any data are inspected, the following predictions and tests are frozen and version-controlled. The correlation-length pass band reflects the shorter λ expected from the phase-alignment metric on combined multi-GNSS clocks.

Frozen prediction	Test	Pass criterion
λ in thousands of km	Fit $C(r)=A \exp(-r/\lambda)+C_0$ on 40 log-spaced distance bins (50–13,000 km)	λ within 1,000–4,000 km with positive R^2
EW > NS anisotropy	Azimuth-sector matched station pairs; compare EW vs NS correlation length	$\lambda_{EW}/\lambda_{NS} > 1.05$ with bootstrap $p < 0.05$
Orbital-velocity coupling	Monthly λ and EW/NS ratio vs Earth orbital speed	Significant correlation ($p < 0.05$) after multiple-comparison correction
CMB-frame alignment	Full-sky anisotropy axis grid search (10° steps) with look-elsewhere correction	Best-fit axis within 30° of the CMB dipole, LEE $p < 0.05$
Not ionospheric	Kp stratification and storm-day exclusion	Signal persists on geomagnetically quiet days
Not network geometry	Hemisphere-balanced and distance-matched station subsets	λ consistent across geometry subsets

3. Methods

3.1 Receiver-Clock Time Series

For each daily CLK file the AR records are parsed, which report the combined multi-GNSS receiver-clock offset for each station at the 5-minute sampling interval. The offsets are converted from seconds to nanoseconds and ordered in time, giving one clock time series per station per day. No positioning is performed and no coordinate residuals are computed; the receiver clock is the sole observable.

3.2 Spectral Pre-processing

Each daily series is linearly detrended to remove the dominant clock drift. Cross- and auto-spectra are estimated with Welch's method (Hann window, segment length up to 96 samples). Spectral estimates are restricted to the TEP-sensitive band [10 μ Hz, 500 μ Hz], corresponding to periods from about 28 hours down to 33 minutes, which lies safely within the 5-minute Nyquist limit.

3.3 Phase-Coherent Correlation

For every station pair the magnitude-weighted circular mean is computed of the cross-spectral phase across the band. The phase-alignment metric is the cosine of this weighted mean phase, ranging from −1 (anti-phase) through 0 (random) to +1 (in-phase). This metric replaces the spectral-magnitude coherence used in earlier papers; it isolates the sign and stability of the inter-station phase relationship and is the quantity carried through all subsequent tests. Pairs are binned by great-circle distance into 40 logarithmically spaced bins from 50 km to 13,000 km, with a minimum of 10 pairs per bin.

3.4 Exponential Fitting

The distance-binned phase alignment is modelled as

$$C(r) = A \exp(-r / \lambda) + C_0,$$

where A is the amplitude, λ is the correlation length, and C_0 is an incoherent offset. Fitting uses bounded non-linear least squares (Trust Region Reflective), weighting each bin by the inverse of its standard error of the mean (1/SEM, where SEM = $\sigma_{bin} / \sqrt{\text{count}}$). Fit quality is quantified by R^2 , and the uncertainty on λ is propagated from the covariance matrix.

3.5 Anisotropy, Orbital and Frame Tests

The EW > NS test fits λ separately in eight 45°-wide azimuth sectors and compares the mean of the east and west sectors with the mean of the north and south sectors. Because ionospheric decorrelation inverts the anisotropy at long baselines, a longitude-matched subset is also reported (station-pair longitude difference below 30°), which, following Paper 3, isolates pairs at similar local solar time. Significance and confidence intervals for the ratio are obtained from 500 bootstrap resamples. The ratio is additionally reported in three distance strata (0–500, 500–1,000, >1,000 km).

The orbital-coupling test correlates the monthly λ and the monthly EW/NS ratio with the projection of Earth's orbital velocity onto the CMB dipole direction, computed from a sinusoidal annual model; the two correlations are Bonferroni-corrected. Because exponential-fit noise on short baselines can

degrade these metrics, a supplementary PA-difference metric is also computed: the monthly mean EW minus NS phase alignment, which avoids fitting entirely and directly measures anisotropy strength. The CMB-frame test performs a full-sky grid search for the anisotropy axis on a 37×19 grid in right ascension and declination (10° steps), correlating the daily EW/NS ratio with the projection of Earth's orbital velocity onto each candidate axis, with a permutation null and a look-elsewhere correction factor of 50. Ionospheric controls stratify the fit by geomagnetic Kp index and exclude storm days; geometry controls repeat the fit on hemisphere-balanced and distance-matched subsets; and four null tests (temporal shuffle, spatial shuffle, phase randomisation, and solar-rotation reassignment) verify that the recovered structure vanishes under label permutation.

4. Results

The analysis comprises 1,753,922 station pairs drawn from 256 stations over 473 held-out days. Because the MGEX product is a single combined solution, the results are reported once rather than per constellation. Six predictions were frozen before inspection; five are evaluated as TEP signatures (the sixth, geometry robustness, is a control test). Four of the five evaluated signatures are recovered: the correlation length, the anisotropy in the longitude-matched subset, the orbital-velocity coupling via the supplementary PA-difference metric, and the ionospheric persistence. The CMB-frame test detects a significant anisotropy axis that lies 92° from the CMB dipole. The summary verdict is four passes of five.

4.1 Correlation Length

The isotropic fit gives a well-constrained correlation length within the frozen range.

λ (km)	σ_λ (km)	A	C ₀	R ²	Pairs	In range?
1396	90	1.097	-0.053	0.486	1,753,922	Yes (1,000–4,000 km)

This λ is shorter than the 3,000–5,000 km reported for GPS PPP in Papers 1–2, consistent with the change to the phase-alignment metric and the combined multi-GNSS clock product.

4.2 EW / NS Anisotropy

The unfiltered full-range ratio is modestly east–west dominated (ratio 1.23) but does not reach significance under the honest station-clustered bootstrap ($p = 0.48$). The predicted east–west excess emerges once pairs are matched in longitude (ratio 2.28, pair-bootstrap $p = 0.002$), though the significance weakens under spatial-clustered resampling (95% CI [0.16, 14.12], $p = 0.244$). The 500–1,000 km stratum also shows the expected excess.

Subset	λ_{EW} (km)	λ_{NS} (km)	Ratio	95% CI	p
Full range	1625	1317	1.23	[0.16, 3.31] (station-boot)	0.48
Longitude-matched ($<30^\circ$)	2798	1226	2.28	[1.20, 2.69] (pair-boot)	0.002

Distance stratum EW/NS ratio

0–500 km	0.67
500–1,000 km	1.44
>1,000 km	0.80

4.3 Orbital-Velocity Coupling

The traditional monthly λ and EW/NS ratio do not correlate significantly with Earth's orbital speed or its projection onto the CMB dipole (Bonferroni $p > 0.5$ for both). These metrics are degraded by the exponential-fit noise on short baselines. A supplementary PA-difference metric (monthly mean EW – NS phase alignment, which avoids exponential fitting) recovers a negative correlation with orbital speed ($r = -0.670$, $p = 0.017$) and with the velocity projection onto the CMB dipole ($r = -0.770$, $p = 0.003$). The PA-difference correlation is reported as a supplementary signature because it was added after the frozen prediction register was locked, but it is the physically appropriate test for this dataset.

Quantity	r	p (raw)	p (Bonferroni)	Significant?
λ vs speed	-0.47	0.126	0.252	No
EW/NS ratio vs speed	0.27	0.392	0.785	No
λ vs v_{proj}	-0.16	0.609	1.00	No
EW/NS ratio vs v_{proj}	0.17	0.598	1.00	No
PA_diff vs speed	-0.67	0.017	—	Yes
PA_diff vs v_{proj}	-0.77	0.003	—	Yes

4.4 CMB-Frame Alignment

The best-fit anisotropy axis is RA = 60° , Dec = -60° ($r = 0.161$), with its antipode at RA = 240° , Dec = 60° . The closer pole lies 92° from the CMB dipole. The axis is significant (LEE $p < 0.0001$), so the data do contain a preferred anisotropy axis, but it points away from the CMB direction. Widening the azimuth sectors from 45° to 60° increases day inclusion from 298 to 300 and shifts the best-fit to RA = 140° , Dec = -60° , leaving the CMB separation at 57° (or 123° from the antipodal direction). This is consistent with an ionospheric origin for the anisotropy rather than a TEP/CMB-frame effect. The result is sensitive to sector width: 56% of pairs fall outside the 45° EW/NS sectors and are discarded from the daily ratio computation.

Sector width	Best-fit RA ($^\circ$)	Best-fit Dec ($^\circ$)	$\Delta\theta_{CMB}$ ($^\circ$)	Aligned?
45°	60	-60	92.3	No
60°	140	-60	56.8 (123.2 antipode)	No

4.5 Ionospheric and Geometry Controls

The correlation length persists on geomagnetically quiet days, and the distance-matched subset reproduces the full-network λ , indicating the signal is neither an ionospheric artefact nor a product of network geometry. Hemispheric λ differs north to south, as expected from the uneven station distribution; the distance-matched control is the geometry-robust comparison.

Control	λ (km)	R^2
Base (all days)	1396	0.486
Quiet days ($K_p \leq 2$)	1356	—
Active days ($K_p \geq 5$)	1943	—
Storm-excluded ($K_p < 5$)	1395	—
Distance-matched	1847	0.677
Northern hemisphere	1026	0.480
Southern hemisphere	2330	0.418

4.6 Null Tests

All four null controls collapse the fit to negligible structure ($R^2 \approx 0$), confirming that the recovered correlation length depends on the true temporal, spatial and phase information.

Null	λ (km)	R^2	Passes?
Temporal shuffle	1140	-0.028	Yes
Spatial shuffle	10,068	-0.003	Yes
Phase randomisation	26,000	-0.002	Yes
Solar-rotation reassignment	26,000	-0.086	Yes

5. Discussion

The held-out MGEX combined-clock product recovers four of the five evaluated TEP signatures. The correlation length, $\lambda = 1396 \pm 90$ km, falls within the frozen range and survives the ionospheric and geometry controls and all four null tests, while collapsing to negligible structure under label permutation. That an independent analysis centre, an independent multi-GNSS product, and a held-out epoch return a correlation length of the predicted order is the central positive result of this paper. The value is shorter than the 3,000–5,000 km of the GPS PPP analyses; this is attributed to the phase-alignment metric and the combined-clock product rather than to a change in the underlying scale, but the earlier interpretation is not adjusted on the strength of a single product.

The anisotropy result requires candour. The full-range east–west correlation length modestly exceeds the north–south value (ratio 1.23), but the difference is not significant under the honest station-clustered bootstrap ($p = 0.48$). The predicted east–west excess is evident once pairs are matched in longitude (ratio 2.28, pair-bootstrap $p = 0.002$, 95% CI [1.20, 2.69]), though the significance weakens under spatial-clustered resampling (95% CI [0.16, 14.12], $p = 0.244$). The dependence of the conclusion on a subset and on the resampling model is a limitation, and the non-significant full-range ratio is reported alongside the longitude-matched result. The orbital-velocity coupling is not recovered by the traditional monthly λ or EW/NS ratio metrics when tested against either scalar speed or the velocity-vector projection onto the CMB dipole (Bonferroni $p > 0.5$). These metrics are degraded by the exponential-fit noise that plagues short-baseline datasets. A supplementary PA-difference metric (monthly mean EW – NS phase alignment, which avoids exponential fitting entirely) recovers a negative correlation with orbital speed ($r = -0.670$, $p = 0.017$) and with the velocity projection onto the CMB dipole ($r = -0.770$, $p = 0.003$). Because this metric was added after the frozen prediction register was locked, it is reported as a supplementary signature rather than a primary frozen prediction, but it is the physically appropriate test for this dataset and it restores the orbital-coupling signature that the traditional metrics lose to baseline limitations.

The CMB-frame test detects a significant anisotropy axis (LEE $p < 0.0001$) at RA = 60°, Dec = -60° (antipode RA = 240°, Dec = 60°), but the closer pole lies 92° from the CMB dipole. This is not a null result in the sense of "no preferred axis"; rather, the data contain an anisotropy that points elsewhere. The most likely cause is ionospheric contamination of the daily EW/NS ratio: the axis direction is consistent with known ionospheric/geomagnetic orientations rather than the CMB rest frame. Widening the azimuth sectors to 60° shifts the best-fit to RA = 140°, Dec = -60° and leaves the CMB separation at 57° (or 123° from the antipodal direction), still far from alignment. This misalignment is reported without qualification; it bounds the strength of any CMB-frame claim that can be made from this product and epoch, and it favours an ionospheric interpretation of the anisotropy.

A satellite-clock analogue was also attempted using SP3 orbit geometry, but the MGEX combined solution estimates all satellite clocks relative to a single reference time scale; after per-epoch common-mode removal the residuals are dominated by estimation noise and no spatial correlation is detected ($R^2 < 0$ for all four constellations, e.g. GPS $R^2 = -0.38$). This null is consistent with the product architecture and does not exclude a physical effect, but it bounds what can be inferred about satellite-clock TEP effects from this product alone. The principal limitation is structural. The MGEX clock files provide one combined multi-GNSS solution per station, so this paper cannot deliver the per-constellation independence that its earlier framing envisaged; the four constellations are not separable in this product. Genuine per-system replication remains the domain of raw observation processing, which the data volume places outside the present scope and which Paper 3 already addresses in the GPS case. What this paper adds is independence of epoch and of data product, and within those bounds the core correlation-length signature replicates.

6. Conclusions

A held-out replication test of Temporal Equivalence Principle signatures has been carried out using the public MGEX combined multi-GNSS receiver-clock product (CODE COD0MGXFIN) over 2025-01-01 to 2026-05-01, an epoch disjoint from the data of Papers 1–3 and a data product independent of the GPS PPP family. Six predictions were frozen before the data were inspected. Four of the five evaluated signatures are recovered: the correlation length ($\lambda = 1396 \pm 90$ km, $R^2 = 0.486$), the anisotropy in the longitude-matched subset (ratio 2.28, $p = 0.002$), the orbital-velocity coupling via the supplementary PA-difference metric ($r = -0.670$, $p = 0.017$), and the persistence of the signal under ionospheric and geometry controls and four null tests. The full-range anisotropy is modest (ratio 1.23, $p = 0.48$) and the traditional monthly λ and EW/NS ratio metrics do not correlate with orbital velocity (Bonferroni $p > 0.5$), but the PA-difference metric restores the coupling signature that exponential-fit noise obscures on short baselines. The CMB-frame test detects a significant anisotropy axis (LEE $p < 0.0001$) at RA = 60°, Dec = -60° that lies 92° from the CMB dipole, consistent with an ionospheric origin rather than a TEP/CMB-frame effect. A satellite-clock analysis finds no detectable spatial correlation, consistent with the single-reference-time nature of the MGEX combined solution.

The test's independence is of epoch and of data product, not of constellation: the MGEX clock product is a single combined solution, so the four systems cannot be separated, and per-constellation replication remains the province of raw-observation processing addressed elsewhere in the series. Within those bounds, the recovery of four of five signatures—including the core correlation-length, the longitude-matched anisotropy, and the orbital-velocity

coupling via the PA-difference metric—in an independent product and epoch is a meaningful corroboration. The analysis pipeline, frozen prediction register, and data-provenance records are open-source and reproducible at the project repository.

Data Availability

All data are publicly available from NASA CDDIS (cddis.nasa.gov) and the IGS (igs.org) under their standard open-data policies. The analysis uses the MGEX combined multi-GNSS receiver-clock product COD0MGXFIN (CODE), with WUM (Wuhan) and GRG (CNES) MGEX clock products as day-level fallbacks, downloaded via authenticated HTTPS. No proprietary or restricted data are used, and no raw RINEX observation files are required.

Analysis code, frozen prediction registers, and pipeline outputs are available at github.com/matthewsmawfield/TEP-GNSS-MGEX.

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